A Fourth-Order Symmetric Composition of the Implicit Projected ABBA Method for Non-Autonomous GC2D Systems

Abstract

This note combines two constructions. First, a non-autonomous second-order ABBA step is made implicit by Hairer's symmetric projection on the duplicated phase space. The projection multiplier is determined from a reduced two-dimensional nonlinear system, and its exact Jacobian can be written explicitly. Second, three complete implicit projected ABBA steps are composed with the symmetric coefficients γ, δ, γ in order to cancel the complete third-order defect. The resulting method has local truncation error $\mathcal{O}(h^5)$ and global order four. Each outer stage has its own multiplier and its own signed time interval, so one fourth-order step requires three complete reduced implicit solves.

1 GC2D system and duplicated phase space

The physical state is

$$z = (X, Y) \in \mathbb{R}^2, \tag{1}$$

and the non-autonomous guiding-center dynamics is written as

$$\dot{z} = f(z,t), \qquad f(z,t) = J_0 \nabla_z \phi(z,t), \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{2}$$

Its spatial Jacobian is

$$W(z,t) := D_z f(z,t) = J_0 \nabla_z^2 \phi(z,t). \tag{3}$$

As in the implicit ABBA construction, duplicate the state:

$$Y = (u, v) \in \mathbb{R}^4, \qquad u, v \in \mathbb{R}^2. \tag{4}$$

The physical state is represented by the diagonal manifold

$$\mathcal{M} = \{(u,v) \in \mathbb{R}^4 : \ u = v\}. \tag{5}$$

Introduce

$$G = \begin{pmatrix} I_2 & -I_2 \end{pmatrix}, \qquad E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}, \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}. \tag{6}$$

Hence

$$Ez = (z,z), \qquad N\mu = (\mu, -\mu), \qquad GG^T = 2I_2. \tag{7}$$

2 The basic non-autonomous ABBA map for an arbitrary signed step

The key point for the fourth-order composition is that we need the same ABBA routine for an arbitrary *signed* substep length ℓ. Let the substep start at time τ and finish at

$$\tau^+ = \tau + \ell. \tag{8}$$

Define

$$s = \frac{\ell}{2}. \tag{9}$$

The four explicit duplicated-space stages are

$$u^{(1)} = u^{(0)} + sf\big(v^{(0)}, \tau\big), \tag{10}$$

$$v^{(1)} = v^{(0)} + sf\big(u^{(1)}, \tau\big), \tag{11}$$

$$v^{(2)} = v^{(1)} + sf\big(u^{(1)}, \tau^+\big), \tag{12}$$

$$u^{(2)} = u^{(1)} + sf\big(v^{(2)}, \tau^+\big). \tag{13}$$

Thus

$$\Phi^{\mathrm{ABBA}}_{\ell,\tau} = A_{s,\tau^+} \circ B_{s,\tau^+} \circ B_{s,\tau} \circ A_{s,\tau}. \tag{14}$$

The two endpoint times τ and τ^+ are essential. In particular, if $\ell < 0$, then $\tau^+ < \tau$ and the same formulas automatically describe a backward non-autonomous ABBA step.

3 One complete implicit projected ABBA step

We now define the actual second-order physical map used inside the new fourth-order method. Given a physical state z at time τ, introduce a multiplier

$$\mu \in \mathbb{R}^2 \tag{15}$$

and pre-correct the duplicated input by

$$Y^{(0)}(\mu) = Ez + N\mu = \begin{pmatrix} z+\mu \\ z-\mu \end{pmatrix}. \tag{16}$$

Therefore

$$u^{(0)} = z + \mu, \qquad v^{(0)} = z - \mu. \tag{17}$$

Apply the four stages (10)–(13) and denote the auxiliary output by

$$\widetilde{Y}^+(\mu) = \Phi^{\mathrm{ABBA}}_{\ell,\tau}(Ez + N\mu) = \begin{pmatrix} u^{(2)}(\mu) \\ v^{(2)}(\mu) \end{pmatrix}. \tag{18}$$

The same multiplier is then applied at the output:

$$Y^+(\mu) = \widetilde{Y}^+(\mu) + N\mu = \begin{pmatrix} u^{(2)}(\mu) + \mu \\ v^{(2)}(\mu) - \mu \end{pmatrix}. \tag{19}$$

We impose $Y^+(\mu) \in \mathcal{M}$. Since G measures the difference between the two copies, the multiplier is determined by the reduced residual

$$\boxed{R_{\ell,\tau}(z,\mu) := G\Phi^{\mathrm{ABBA}}_{\ell,\tau}(Ez + N\mu) + 2\mu = 0.} \tag{20}$$

Equivalently,

$$R_{\ell,\tau}(z,\mu) = u^{(2)}(z,\mu) - v^{(2)}(z,\mu) + 2\mu. \tag{21}$$

If μ^* is the converged root, the physical output is

$$\boxed{\Psi^{[2]}_{\ell,\tau}(z) = u^{(2)}(\mu^*) + \mu^* = v^{(2)}(\mu^*) - \mu^*.} \tag{22}$$

Thus, in this document,

$$\boxed{S_2(\tau,\ell) \equiv \Psi^{[2]}_{\ell,\tau}} \tag{23}$$

means one *complete* implicit ABBA step: pre-projection, four ABBA stages, nonlinear multiplier solve, and post-projection.

4 Exact reduced Jacobian for the multiplier solve

For a fixed evaluation of the four stages, define

$$W_1 = W(v^{(0)},\tau), \qquad W_2 = W(u^{(1)},\tau), \tag{24}$$

$$W_3 = W(u^{(1)},\tau^+), \qquad W_4 = W(v^{(2)},\tau^+), \tag{25}$$

and

$$S := W_2 + W_3. \tag{26}$$

The exact Jacobian of the duplicated ABBA map is

$$J_{\mathrm{ABBA}} := D\Phi^{\mathrm{ABBA}}_{\ell,\tau} = \begin{pmatrix} I_2 + s^2 W_4 S & s(W_1 + W_4) + s^3 W_4 S W_1 \\ sS & I_2 + s^2 S W_1 \end{pmatrix}. \tag{27}$$

No commutation of the matrices W_i is assumed; the product order in (27) must be preserved.

Differentiating (20) with respect to the multiplier gives

$$\boxed{K_{\ell,\tau}(z,\mu) := D_\mu R_{\ell,\tau}(z,\mu) = G J_{\mathrm{ABBA}} N + 2I_2.} \tag{28}$$

Writing the blocks of (27) explicitly, this becomes

$$\boxed{\begin{aligned} K_{\ell,\tau} = 4I_2 &- s(W_1 + W_2 + W_3 + W_4) \\ &+ s^2\left[W_4(W_2 + W_3) + (W_2 + W_3)W_1\right] \\ &- s^3 W_4(W_2 + W_3)W_1. \end{aligned}} \tag{29}$$

Remember that here

$$s = \frac{\ell}{2}, \tag{30}$$

so the same formula is valid for positive and negative outer substeps.

A Newton iteration for the k-th implicit ABBA solve is therefore

$$\boxed{\mu^{(j+1)} = \mu^{(j)} - K_{\ell,\tau}(z,\mu^{(j)})^{-1} R_{\ell,\tau}(z,\mu^{(j)}).} \tag{31}$$

Because $K \in \mathbb{R}^{2\times 2}$, its inverse can be evaluated explicitly if desired.

5 Exact Jacobian of one implicit physical substep

The reduced Jacobian $K = D_\mu R$ is the matrix needed by Newton, but it is not the Jacobian of the physical map $z \mapsto \Psi^{[2]}_{\ell,\tau}(z)$. The converged multiplier depends implicitly on z.

Write

$$J_{\mathrm{ABBA}} = \begin{pmatrix} \mathsf{A} & \mathsf{B} \\ \mathsf{C} & \mathsf{D} \end{pmatrix}, \tag{32}$$

where

$$\mathsf{A} = I_2 + s^2 W_4 S, \tag{33}$$

$$\mathsf{B} = s(W_1 + W_4) + s^3 W_4 S W_1, \tag{34}$$

$$\mathsf{C} = sS, \tag{35}$$

$$\mathsf{D} = I_2 + s^2 S W_1. \tag{36}$$

Besides K, define

$$L := D_z R = G J_{\mathrm{ABBA}} E = \mathsf{A} + \mathsf{B} - \mathsf{C} - \mathsf{D}. \tag{37}$$

The implicit function theorem gives

$$D_z \mu^*(z) = -K^{-1} L. \tag{38}$$

Define

$$P := \mathsf{A} + \mathsf{B}, \qquad Q := \mathsf{A} - \mathsf{B} + I_2. \tag{39}$$

Then the exact physical Jacobian of one implicit projected ABBA substep is

$$\boxed{J_{\Psi^{[2]}} := D_z \Psi^{[2]}_{\ell,\tau}(z) = P - Q K^{-1} L.} \tag{40}$$

This formula will also give the Jacobian of the final fourth-order method by a simple product of three 2×2 matrices.

6 Fourth-order composition: three complete implicit ABBA solves

Let

$$\boxed{\gamma = \frac{1}{2 - 2^{1/3}}, \qquad \delta = -\frac{2^{1/3}}{2 - 2^{1/3}}.} \tag{41}$$

Numerically,

$$\gamma \approx 1.35120719196, \qquad \delta \approx -1.70241438392. \tag{42}$$

They satisfy

$$2\gamma + \delta = 1, \qquad 2\gamma^3 + \delta^3 = 0. \tag{43}$$

Starting from (t_n, z_n), define the three signed substeps

$$\ell_1 = \gamma h, \qquad \ell_2 = \delta h, \qquad \ell_3 = \gamma h, \tag{44}$$

and their starting times

$$\tau_1 = t_n, \qquad \tau_2 = t_n + \gamma h, \qquad \tau_3 = t_n + (\gamma + \delta) h. \tag{45}$$

Then the new fourth-order map is

$$\boxed{\Psi^{[4]}_{h,t_n} = \Psi^{[2]}_{\ell_3,\tau_3} \circ \Psi^{[2]}_{\ell_2,\tau_2} \circ \Psi^{[2]}_{\ell_1,\tau_1}.} \tag{46}$$

Equivalently,

$$z^{[1]} = \Psi^{[2]}_{\gamma h, t_n}(z_n), \tag{47}$$

$$z^{[2]} = \Psi^{[2]}_{\delta h, t_n+\gamma h}(z^{[1]}), \tag{48}$$

$$z_{n+1} = \Psi^{[2]}_{\gamma h, t_n+(\gamma+\delta)h}(z^{[2]}). \tag{49}$$

Each line (47)–(49) contains one separate nonlinear multiplier solve.

The three multipliers

The three roots should be distinguished explicitly:

$$R_{\gamma h, t_n}(z_n, \mu_1) = 0, \tag{50}$$

$$R_{\delta h, t_n+\gamma h}(z^{[1]}, \mu_2) = 0, \tag{51}$$

$$R_{\gamma h, t_n+(\gamma+\delta)h}(z^{[2]}, \mu_3) = 0. \tag{52}$$

Thus a single fourth-order step contains

$$\boxed{\mu_1,\ \mu_2,\ \mu_3,} \tag{53}$$

not one common multiplier. Each multiplier belongs to one complete implicit projected ABBA substep.

7 Time structure inside the three implicit ABBA substeps

The non-autonomous endpoint evaluations must be recomputed for every outer substep.

For the first solve,

$$\tau_1 = t_n, \qquad \tau_1^+ = t_n + \gamma h. \tag{54}$$

Therefore its internal ABBA pattern is

$$A_{\gamma h/2, t_n} \to B_{\gamma h/2, t_n} \to B_{\gamma h/2, t_n+\gamma h} \to A_{\gamma h/2, t_n+\gamma h}. \tag{55}$$

For the second solve,

$$\tau_2 = t_n + \gamma h, \qquad \tau_2^+ = t_n + (\gamma+\delta)h. \tag{56}$$

Since $\delta < 0$, the internal half-step is

$$s_2 = \frac{\delta h}{2} < 0, \tag{57}$$

and this complete ABBA step evolves backward from τ_2 to τ_2^+.

For the third solve,

$$\tau_3 = t_n + (\gamma+\delta)h, \qquad \tau_3^+ = t_n + h. \tag{58}$$

Because $2\gamma + \delta = 1$, the complete composition ends exactly at

$$t_{n+1} = t_n + h. \tag{59}$$

8 Detailed implementation algorithm

One complete fourth-order step can be implemented as follows.

1. Set the outer coefficients $c_1 = \gamma$, $c_2 = \delta$, $c_3 = \gamma$.

2. Initialize

$$z^{[0]} = z_n, \qquad \tau_1 = t_n. \tag{60}$$

3. For $k = 1, 2, 3$, set

$$\ell_k = c_k h, \qquad s_k = \frac{\ell_k}{2}, \qquad \tau_k^+ = \tau_k + \ell_k. \tag{61}$$

4. For a current Newton iterate $\mu_k^{(j)}$, set

$$u^{(0)} = z^{[k-1]} + \mu_k^{(j)}, \qquad v^{(0)} = z^{[k-1]} - \mu_k^{(j)}. \tag{62}$$

5. Evaluate the four non-autonomous ABBA stages with the signed half-step s_k:

$$\begin{aligned}
u^{(1)} &= u^{(0)} + s_k f(v^{(0)}, \tau_k), \\
v^{(1)} &= v^{(0)} + s_k f(u^{(1)}, \tau_k), \\
v^{(2)} &= v^{(1)} + s_k f(u^{(1)}, \tau_k^+), \\
u^{(2)} &= u^{(1)} + s_k f(v^{(2)}, \tau_k^+).
\end{aligned}$$

6. Form the residual

$$R_k = u^{(2)} - v^{(2)} + 2\mu_k^{(j)}. \tag{63}$$

7. Evaluate W_1, W_2, W_3, W_4 at the four stage points and construct the exact reduced Jacobian K_k from (29), using $s = s_k$.

8. Apply the Newton update

$$\mu_k^{(j+1)} = \mu_k^{(j)} - K_k^{-1} R_k \tag{64}$$

until the chosen convergence criterion is satisfied.

9. With the converged multiplier μ_k^*, recompute or retain the converged stages and set

$$z^{[k]} = u^{(2)} + \mu_k^* = v^{(2)} - \mu_k^*. \tag{65}$$

10. Advance the starting time for the next complete implicit ABBA solve:

$$\tau_{k+1} = \tau_k^+. \tag{66}$$

Finally,

$$\boxed{z_{n+1} = z^{[3]}, \qquad t_{n+1} = t_n + h.} \tag{67}$$

9 Why the implicit composition is fourth order

For the order argument, lift the complete physical second-order step to the extended state (t, z):

$$\widehat{\Psi^{[2]}}_\ell(t, z) := \big(t + \ell, \Psi^{[2]}_{\ell,t}(z)\big). \tag{68}$$

If the multiplier equation is solved exactly and the local root is unique, the symmetric projection together with the palindromic non-autonomous ABBA step gives the reversibility relation

$$\widehat{\Psi^{[2]}}_{-\ell} = \left(\widehat{\Psi^{[2]}}_\ell\right)^{-1}. \tag{69}$$

Hence the logarithm of this complete implicit step has the formal structure

$$\log \widehat{\Psi^{[2]}}_h = h\mathcal{F} + h^3 E_3 + h^5 E_5 + \cdots . \tag{70}$$

Here $\mathcal{F}$ is the vector field of the autonomous extended problem. The element E_3 includes every third-order defect of the complete projected non-autonomous method, including terms generated by the explicit time dependence and by the projection.

Now compose three scaled copies:

$$\widehat{\Psi^{[4]}}_h = \widehat{\Psi^{[2]}}_{\gamma h} \widehat{\Psi^{[2]}}_{\delta h} \widehat{\Psi^{[2]}}_{\gamma h}. \tag{71}$$

The consistency condition gives

$$(2\gamma + \delta)h\mathcal{F} = h\mathcal{F}, \tag{72}$$

and the coefficient of the complete third-order defect is

$$(2\gamma^3 + \delta^3)h^3 E_3 = 0. \tag{73}$$

Because the outer composition is again symmetric, even powers are absent from its logarithm. Therefore

$$\boxed{\log \widehat{\Psi^{[4]}}_h = h\mathcal{F} + h^5 \widetilde{E}_5 + \mathcal{O}(h^7).} \tag{74}$$

Thus

$$\boxed{\text{local truncation error} = \mathcal{O}(h^5),} \tag{75}$$

and over a fixed time interval

$$\boxed{\text{global error} = \mathcal{O}(h^4).} \tag{76}$$

The method is therefore fourth order. It is not a fifth-order symmetric method; a consistent symmetric one-step method has even global order.

10 Symmetry and symplecticity of the complete fourth-order map

For one complete projected ABBA step, exact solution of the multiplier equation gives

$$\Psi^{[2]}_{-\ell,\tau+\ell} \circ \Psi^{[2]}_{\ell,\tau} = I. \tag{77}$$

The reverse multiplier is $-\mu$. Since the outer coefficient sequence (γ, δ, γ) is palindromic, the fourth-order composition also satisfies

$$\Psi^{[4]}_{-h,t_n+h} = \left(\Psi^{[4]}_{h,t_n}\right)^{-1}. \tag{78}$$

Under the local hypotheses used for the implicit projected ABBA method—in particular, local uniqueness of the root, invertibility of K, and exact solution of the projection equation—each physical second-order map is symplectic:

$$\left(D\Psi^{[2]}_{\ell,\tau}\right)^T J_0 D\Psi^{[2]}_{\ell,\tau} = J_0. \tag{79}$$

A composition of symplectic maps is symplectic, hence

$$\boxed{\left(D\Psi^{[4]}_{h,t_n}\right)^T J_0 D\Psi^{[4]}_{h,t_n} = J_0.} \tag{80}$$

In two physical dimensions this is equivalent to

$$\det D\Psi^{[4]}_{h,t_n} = 1. \tag{81}$$

11 Jacobian of the fourth-order implicit physical step

Let

$$J_1 = D\Psi^{[2]}_{\gamma h,t_n}(z_n), \tag{82}$$

$$J_2 = D\Psi^{[2]}_{\delta h,t_n+\gamma h}(z^{[1]}), \tag{83}$$

$$J_3 = D\Psi^{[2]}_{\gamma h,t_n+(\gamma+\delta)h}(z^{[2]}), \tag{84}$$

where each J_k is computed from the explicit implicit-step formula (40) with its own stage matrices and its own multiplier μ_k^*. The chain rule then gives

$$\boxed{D\Psi^{[4]}_{h,t_n}(z_n) = J_3 J_2 J_1.} \tag{85}$$

This gives an explicit route to the Jacobian of the new fourth-order physical method without differentiating the nonlinear solver iterations themselves: one differentiates the converged implicit map through the implicit function theorem at each of the three outer stages.

12 Practical remarks on the nonlinear solves

- The three multipliers μ_1, μ_2, μ_3 solve three different residual equations. They should not be identified with one another.
- The middle solve uses $\ell_2 = \delta h < 0$. Nothing in the residual or Jacobian formulas has to be changed beyond using the signed value $s_2 = \delta h/2$ and the corresponding endpoint times.
- The formal symmetry and symplecticity statements refer to the exact root of each reduced nonlinear system. With a finite stopping tolerance, these properties are preserved only up to the nonlinear-solve defect.
- To observe the designed fourth-order convergence in practice, the error introduced by the multiplier solves should be smaller than the $\mathcal{O}(h^5)$ local discretization error over the step sizes being tested. A fixed very tight tolerance is often sufficient until the integration error reaches the tolerance floor.
- Newton can use the exact 2×2 matrix K in (29). A quasi-Newton method such as Broyden may instead approximate this same reduced Jacobian, without changing the definition of the implicit numerical map when the residual is solved to convergence.

13 Compact schematic view

The complete construction is

$$\boxed{\begin{array}{l} z_n \\ \xrightarrow[\mu_1]{\text{implicit projected ABBA }(\gamma h)} z^{[1]} \\ \xrightarrow[\mu_2]{\text{implicit projected ABBA }(\delta h)} z^{[2]} \\ \xrightarrow[\mu_3]{\text{implicit projected ABBA }(\gamma h)} z_{n+1}. \end{array}} \tag{86}$$

Inside every arrow one solves

$$R_{\ell,\tau}(z,\mu) = G\Phi^{\mathrm{ABBA}}_{\ell,\tau}(Ez + N\mu) + 2\mu = 0, \tag{87}$$

with

$$D_\mu R = K = GJ_{\mathrm{ABBA}}N + 2I_2, \tag{88}$$

and then returns

$$z^+ = u^{(2)} + \mu = v^{(2)} - \mu. \tag{89}$$

The outer coefficients satisfy

$$2\gamma + \delta = 1, \qquad 2\gamma^3 + \delta^3 = 0, \tag{90}$$

so the complete method has local error $\mathcal{O}(h^5)$ and global order four while retaining the symmetry and symplectic structure of the exact projected substeps.

14 Summary

The new method should not be interpreted as three explicit ABBA maps followed by one projection. Rather, it is the composition of three *complete implicit projected ABBA maps*:

$$\boxed{\Psi^{[4]}_{h,t_n} = \underbrace{\Psi^{[2]}_{\gamma h,t_n}}_{\text{implicit solve for }\mu_1} \quad \underbrace{\Psi^{[2]}_{\delta h,t_n+\gamma h}}_{\text{implicit solve for }\mu_2} \quad \underbrace{\Psi^{[2]}_{\gamma h,t_n+(\gamma+\delta)h}}_{\text{implicit solve for }\mu_3},} \tag{91}$$

where composition is understood in the usual right-to-left sense.

Every second-order map uses the same reduced implicit architecture as the original projected ABBA method, with the only changes being its signed step length and its two local endpoint times. This preserves the non-autonomous time symmetry at the inner level and enables the outer symmetric composition to cancel the complete third-order defect.